Demand Side Policies continued: [Fiscal Policy]

Arriving at an alternative formula for multiplier

When income increases, the following scenarios can take place

- Spend more i.e. MPC [ΔC/ΔY]
- Save more i.e. MPS [ΔS/ΔY]
- Pay more tax i.e. MPT [ΔT/ΔY]
- Import more i.e. MPM [ΔM/ΔY]

MPC + MPS + MPT + MPM = 1

Formula for multiplier = $\frac{1}{1-MPC}$

MPS + MPT + MPM = are leakages

Automatic Stabilisers

Part of fiscal policy that can automatically decrease economic fluctuations without any deliberate changes in policies.

a. unemployment benefits

b. progressive income taxes

During a recession – RGDP falls hence, demand for labour will decrease – therefore, unemployment will increase.

Without unemployment benefits – when unemployment increases – income falls a lot – due to which there is a drastic decrease in consumption and AD.

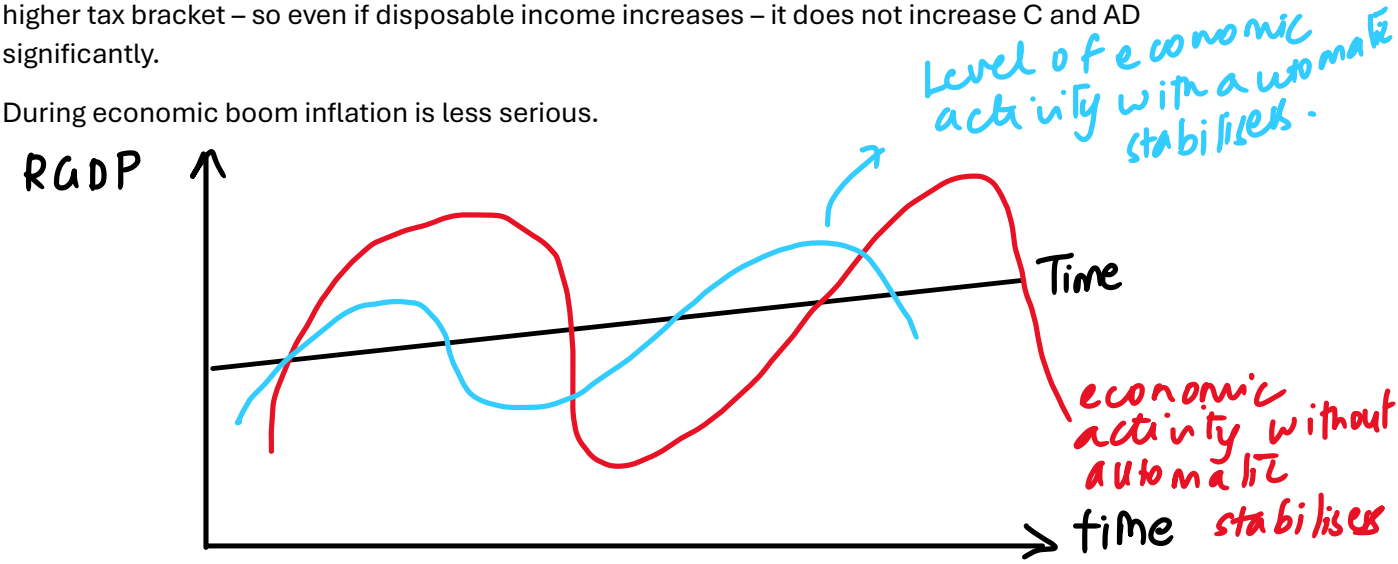
With unemployment benefits – when unemployment increases – people can claim some money automatically hence, the recession is less serious as the fall in C and AD is not drastic.

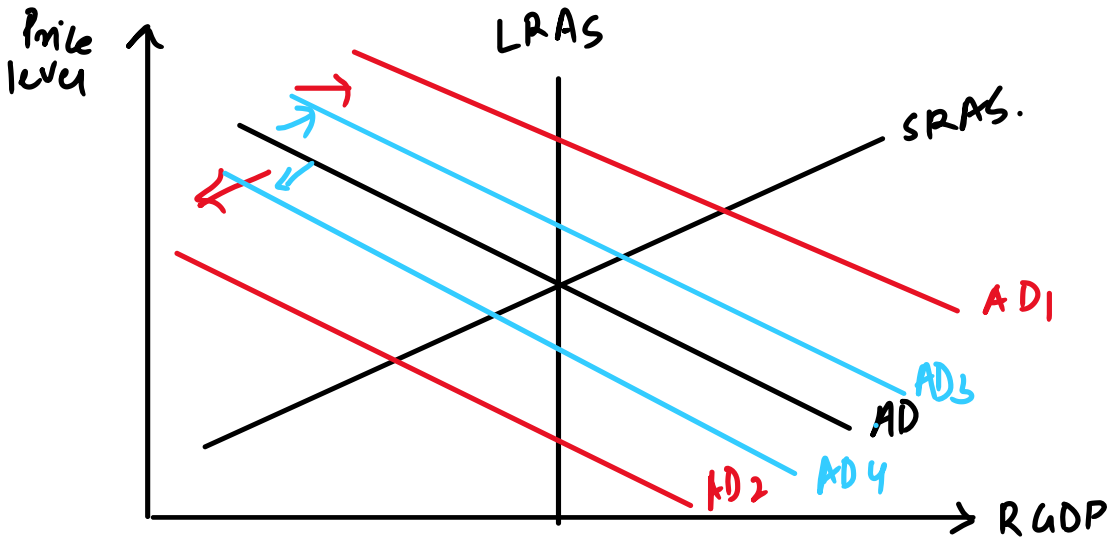
During a Boom – RGDP increases significantly hence, income also increases significantly.

With progressive taxes – when income increases significantly – Consumption also increases a lot hence, the rise in AD is very high.

With progressive taxes – when income increases – income is automatically pushed into the higher tax bracket – so even if disposable income increases – it does not increase C and AD significantly.

During economic boom inflation is less serious.





AD1 & AD2 → without automatic stabilisers

AD3 & AD4 → with automatic stabilisers.

Evaluating Fiscal Policy

Strengths of fiscal Policy

a. Pulling the economy out of deep recession: market forces alone are unable to pull the economy out of deep recession. Wages and Prices are inflexible in downward direction even during steep recession and a fall in AD could keep the economy stuck in a recessionary gap indefinitely.

b. Ability to target sectors of the economy

E.g. education – particular levels of education, healthcare or infrastructure.

c. Direct impact of government spending

When G increases – AD increases

d. Dealing with rapid and escalating inflation

Contractionary fiscal policy – will help decrease AD and Price levels by using high taxation and or low government spending.

e. Ability to affect the potential output

Fiscal policy can create a stable macroeconomic environment – with direct investment in human capital, physical capital which due to increase in firms investment.

f. Automatic Stabilisers

factors that work automatically towards stabilising the economy by reducing short term fluctuations in the business cycle [No need of government action]

a. Progressive Income tax – during growth when RGDP increases – the rate of tax also increases – this is more than increase in income. After tax - disposable income decreases – AD increases less this counteracts the economic expansion making it smaller than it would otherwise be. AD increases slowly.

b. unemployment benefits – during a recession when RGDP falls – unemployment increases – unemployment benefits increase – due to unemployment benefits – C is maintained hence, AD will not be severely low.

Constraints of using fiscal policy

a. Problem of time lags – time lag till inflation is recognised by the government appropriate policy is decided, policy takes time to show effects in the economy.

b. Political constraints

Spending on public goods and merit goods cannot be cut off during contractionary policy. Increase in tax – make the government unpopular – may be avoided even if it is necessary.

c. Sustainable Debt – Government expenditure > government revenue – lead to higher debt

Sustainable debt – refers to a level of debt where a borrowing government can meet its present and future debt obligations without accumulating overdue debt payments.

During recession – debt is unsustainable 0 leading to default inability to pay back.

d. In a recession – tax cuts may not be effective in increasing AD –

low tax – will result in high disposable income – saved due to low confidence about the future.

High government spending – more powerful to increase AD.

e. inability to fine tune the economy – fiscal policy can increase or decrease AD but cannot fine tune the economy – cannot be used to reach a precise target with reference to output, employment and price levels.

f. May be inflationary – If AD increases – Price levels increase – result in demand pull inflation.

g. inability to deal with cost push inflation or stagflation – can only control AD and not SRAS.

Stagflation – increase in inflation and unemployment at the same time.

Monetary Policy

Refers to change in money supply and rate of interest by the central bank to influence the level of AD.

Rate of interest – cost of borrowing or the return on savings.

Higher the rate of interest – borrow less as it expensive to borrow and save more as incentive to save is high.

Money Supply – total volume of money circulating in the economy at a given point of time.

Cash + Bank Deposits [easy to withdraw] + Central Bank Reserves

Types of Monetary Policy

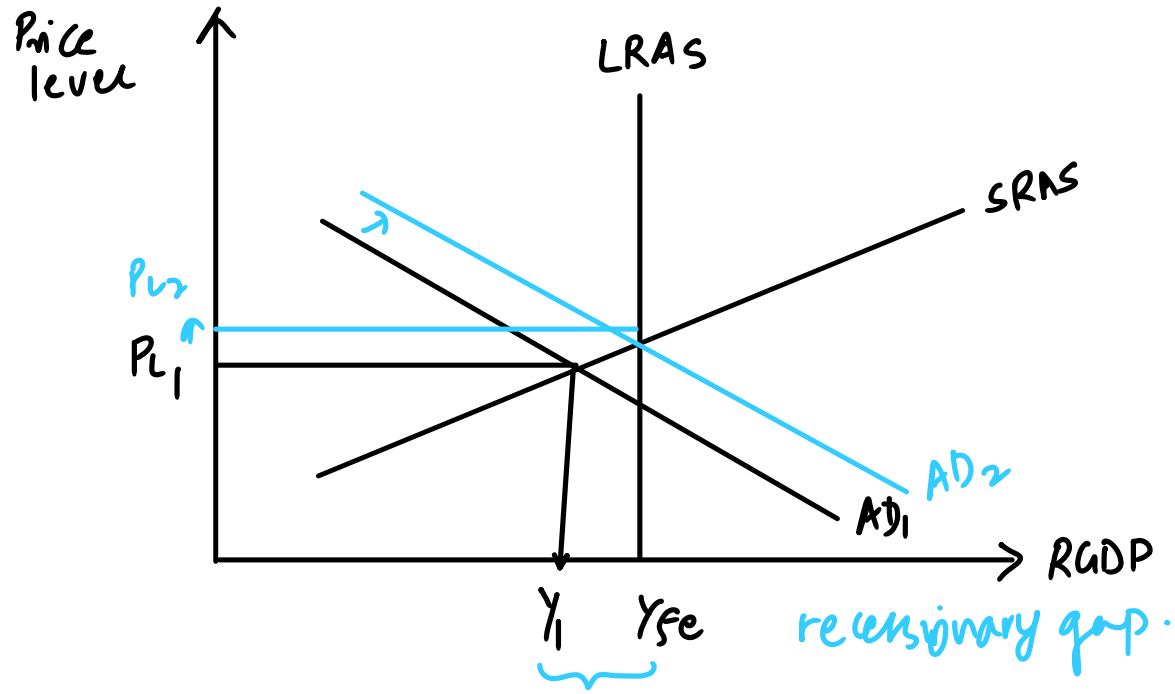
<u>Expansionary Monetary Policy –</u>

Money supply increases – lower rate of interest – lead to higher AD and increase in RGDP – result in economic growth and low unemployment.

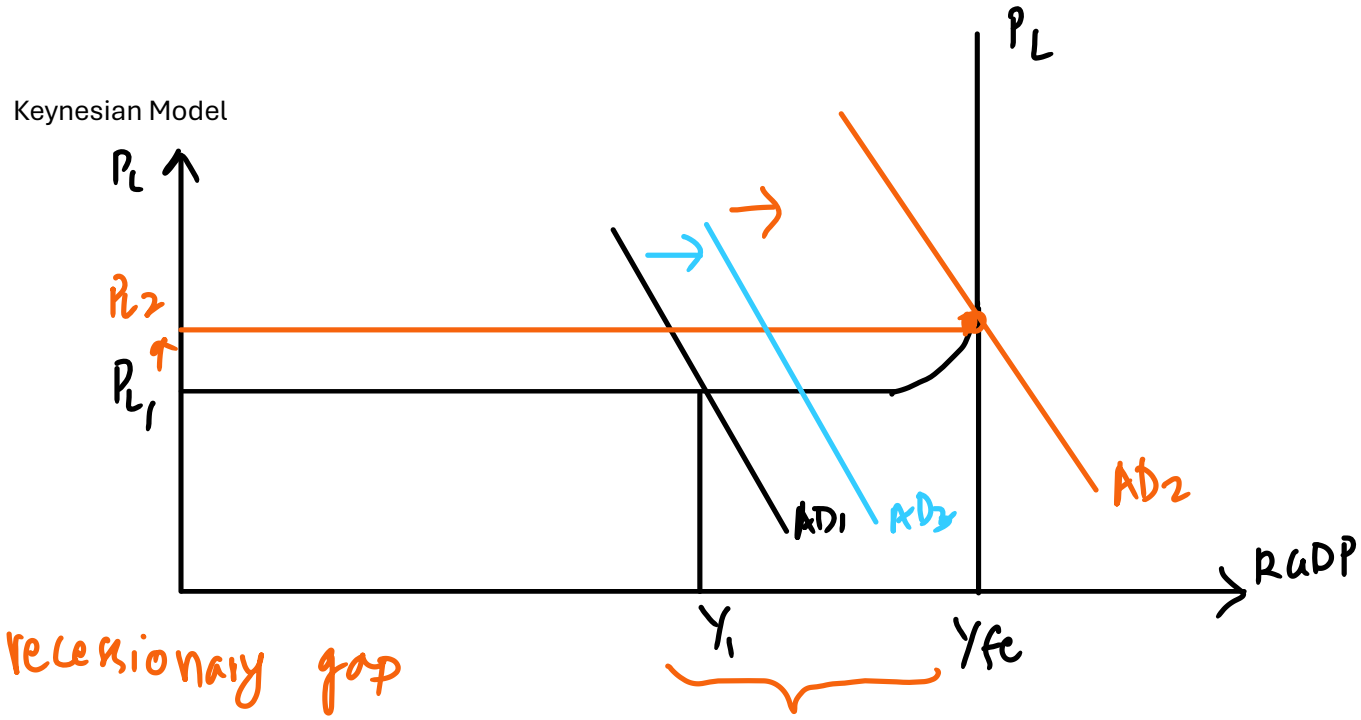
Lower the rate of interest – lower the cost of borrowing – Hence, I will increase – and C will increase – AD will increase – rightward shift – close the recessionary gap.

Due to fall in rate of interest – return on savings is low – households borrow more and spend more.

Monetarist/New Classical Model



Keynesian Model



AD increases – RGDP increases – economic growth increases – demand for labour increases – unemployment will decease – price levels will increase.

If AD increases in the horizontal section – AD1 to AD3 – no change in the price level

If AD increases in the upward sloping part – price levels will increase

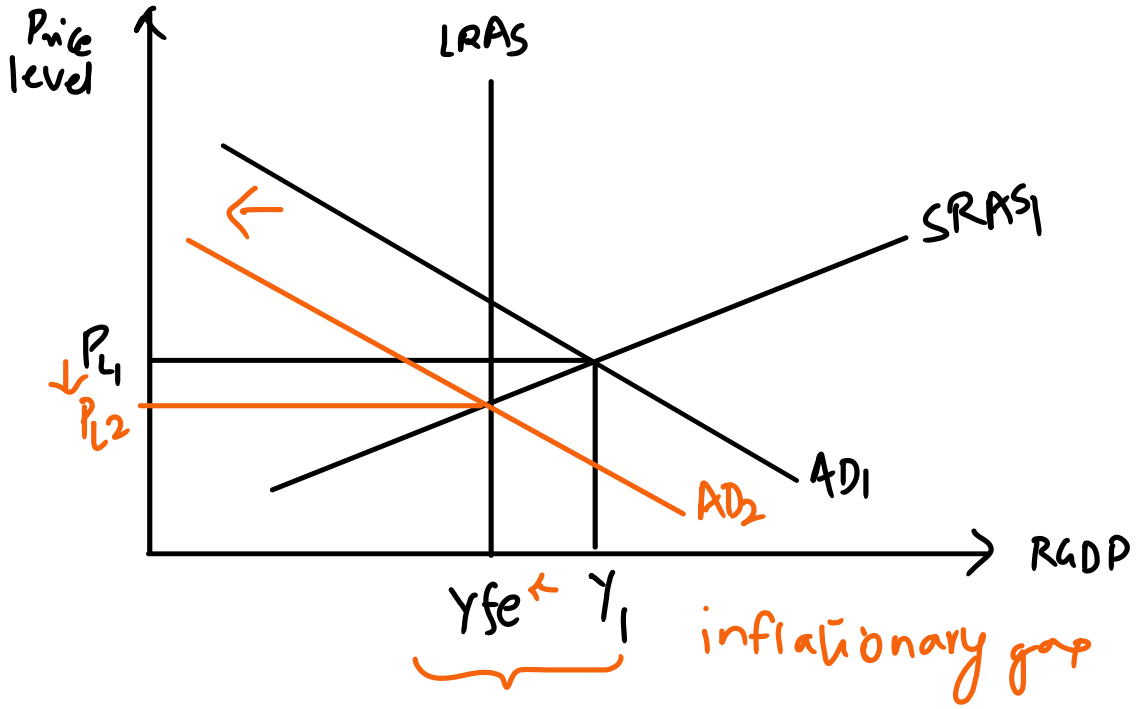
Contractionary Monetary Policy

Money Supply decreases – rate of interest increases – Fall in AD and RGDP hence, price levels fall.

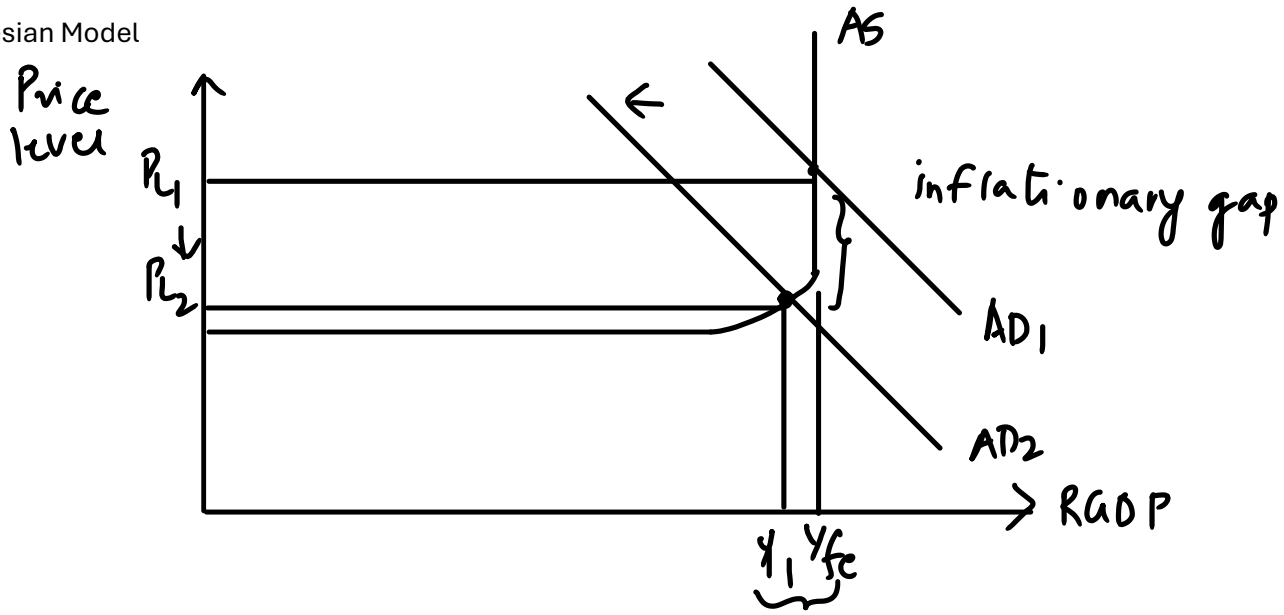
Increase in rate of interest – will increase the cost of borrowing hence, C and I will decrease – AD will decrease – leftward shift – close the inflationary gap.

Due to high rate of interest – savings will increase – and RGDP will fall as C and I will fall.

Monetarist/New Classical Model



Keynesian Model



Inflation targeting

Central banks aim to maintain a particular targeted rate of inflation around 2%.

If inflation rate is 5% it is greater than the targeted rate – use contractionary monetary policy.

If inflation rate is 0.3% it is lower than the targeted rate – use expansionary monetary policy.

Determination of rate of interest

Money market and the rate of interest

Monetary policy uses rate of interest to influence AD

Money – anything that is acceptable as payment for goods and services

Money = Currency notes and coins + deposits that can be withdrawn.

Demand for money and supply of money determine the equilibrium rate of interest.

Supply of money is perfectly inelastic as it is fixed by the central bank

Demand for money

Bonds Vs Money

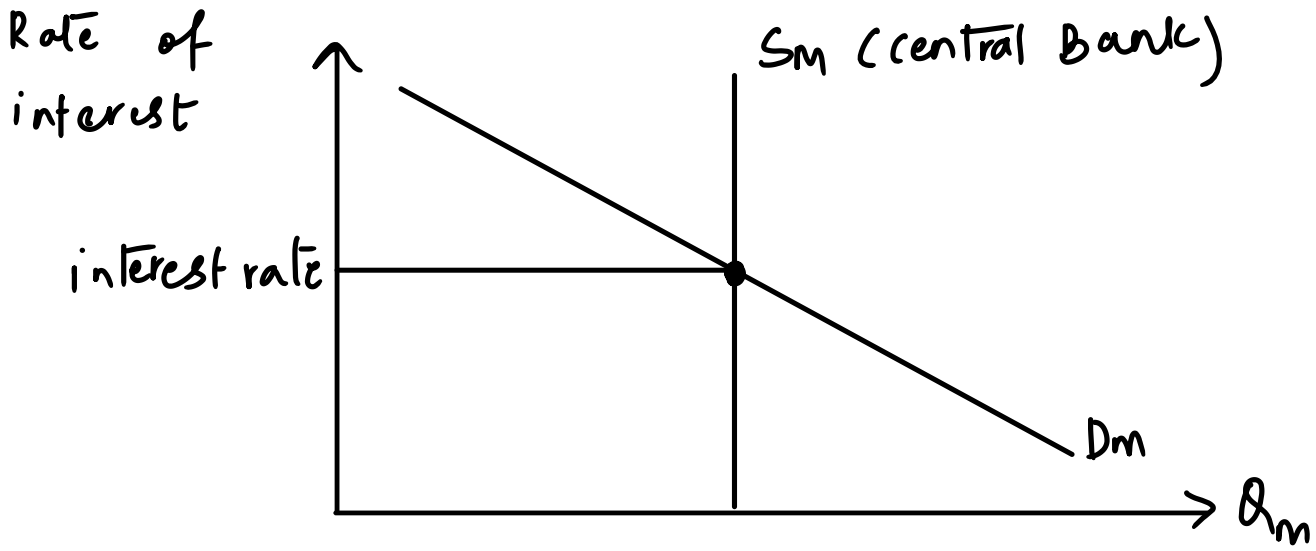
Bonds – no liquidity but there is interest income.

Money – high liquidity but no interest.

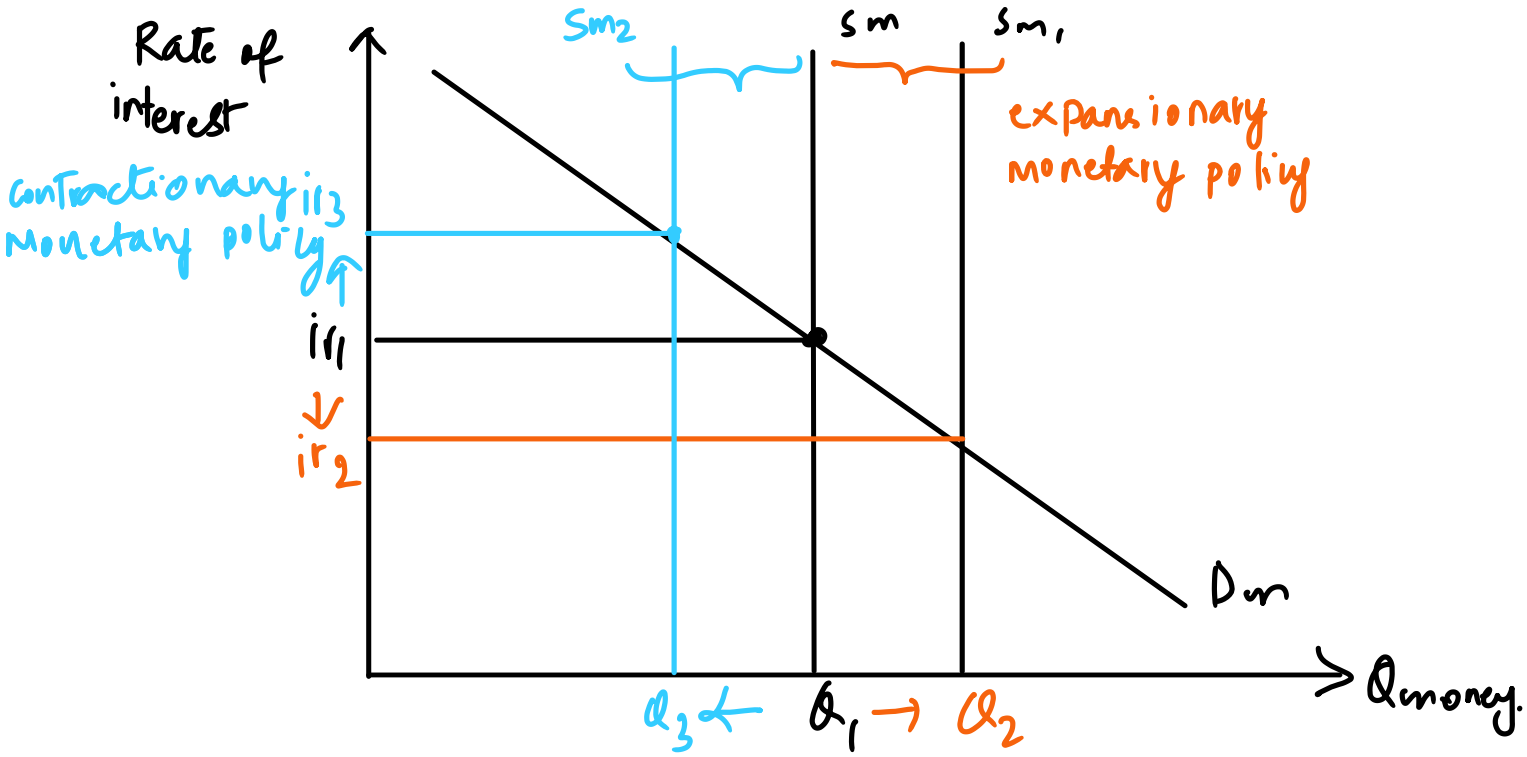
When the rate of interest increases – people would prefer to save in bonds – demand for money decreases.

Rate of interest and demand for money are inversely related.

Determination of rate of interest



Changes in Supply of money – causes changes in equilibrium rate of interest



How central banks change money supply?

1. Open Market Operations [OMO] – central banks buy and sell pre-existing bonds of the private sector in the bond market. The bond market is a simply a market where holders of bonds can buy and sell pre-existing bonds.

Central bank – buy bonds – give money to commercial banks – money supply will increase – rate of interest will decrease.

Central bank – sell bonds – take away money from commercial banks – money supply will decrease - rate of interest will increase.

2. Minimum Reserve Requirement – Refer to that part of bank deposits that commercial keep as required reserves and cannot be loaned out.

Central bank decreases the minimum reserve requirement – commercial banks have excess reserves – ability to lend increase – ability to create money will increase – money supply will increase.

3. Changes in central banks minimum lending rate:

Rate of interest charged by central banks when giving loans to commercial banks.

EU – refinancing rate, US – base rate/discounting rate

If central banks – increase the minimum lending rate – borrowing will become expensive for commercial banks – borrowing will decrease – lending ability will also decrease – money supply will decrease.

4. Quantitate Easing – buying long dated bonds by central bank on much larger.

Objective is to encourage borrowing by firms and consumers in order to increase aggregate demand.

To pay for assets – central bank creates electronic reserves – commercial banks sell the assets – end with more reserves – which can be used for loans.

Next class – credit creating and money multiplier

Homework – QB 6057 – 10, 14